

Spectral-Activity Detection and Stereo Re-Panning of Multiphonic Recordings

Conceptual notes for a center-de-crushing enhancement tool

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1 Objective

- **Input:** an existing stereo recording of an orchestra (multiphonic music).
- **Output:** an enhanced stereo recording in which the instruments are better spread across the left/right field.
- **No target instrument a priori.** The tool detects the spectral activities present in the recording, presents them to the user, and the user assigns each a left/right position.
- **Typical case:** a recording whose instruments are “crushed” to the center, which the transformation moves apart.

2 Signal model

Working observation: the same instrument appears in both channels with varied spectral strength but a nearly identical spectrum (except in the higher frequencies). In the short-time Fourier transform (STFT) domain the mixture is then approximately

$$L(t, f) \approx \sum_i a_i(t) s_i(f) e^{j\phi_i^L(t, f)}, \quad R(t, f) \approx \sum_i b_i(t) s_i(f) e^{j\phi_i^R(t, f)}$$

where

- $s_i(f)$ is the *common source spectrum* (timbre) of activity i ,
- $a_i(t)$, $b_i(t)$ are time-varying per-channel levels,
- ϕ_i^L , ϕ_i^R are per-channel phases.

Two consequences follow immediately:

1. **Timbre is channel-invariant.** Clustering should be done in a single *shared* timbre space spanning both channels, which is far more stable than analyzing each channel separately.

2. **The per-bin L/R ratio is a spatial sample.** Where a single activity i dominates a time–frequency bin,

$$\left| \frac{L}{R} \right| \approx \frac{a_i(t)}{b_i(t)}, \quad \angle \frac{L}{R} \approx \phi_i^L - \phi_i^R,$$

i.e. the inter-channel level difference (ILD) and inter-channel phase difference (ICPD) of the dominant source. Every bin is thus a small measurement of where a source sits.

3 Stage 1 — Detection (what is there)

1. STFT both channels.
2. Enforce the shared-spectrum prior directly: non-negative matrix factorization of the magnitudes (Euclidean objective) with a *common* timbre matrix and *separate* activations per channel,

$$\log |L(t, f)| \approx D A_L, \quad \log |R(t, f)| \approx D A_R.$$

Each column of D is a candidate *activity* (a timbre); A_L, A_R give its time-varying presence per channel.

3. Per activity, compute a **spatial fingerprint**: the median (and histogram) of $|L/R|$ and $\angle(L/R)$ over the bins assigned to it. This shows the user where the activity *actually* sits in the original recording.
4. Weight timbre similarity toward the low/mid bands: high frequencies are more directional, so they are a poor timbre discriminator.
5. Present to the user, per activity: (a) the average spectrum, (b) the time-activation curve, (c) the spatial fingerprint. The user names the activity (“violins”) and assigns a target pan.

Naming note: what is detected are *timbral activities*, not instruments. One instrument playing different notes may split into several activities; the user’s labeling step absorbs this.

4 Stage 2 — Rendering (where to put them)

1. Soft mask per activity $\Rightarrow$ stem extraction. Phase is inherited from the mixture, so masking is phase-safe (no re-synthesis artifacts).
2. **Phase decision — preserve or re-render:**
 - *Preserve:* per-stem, per-channel gain only (change ILD, keep the original ICPD). Safe, but a source cannot be moved far without comb-filtering.
 - *Re-render (recommended for de-crushing):* collapse each stem to mono and render to L/R with the target pan θ using equal-power gains,

$$g_L = \cos \theta, \quad g_R = \sin \theta,$$

with an optional small inter-channel delay for width. Clean control; the original’s minimal spatial detail is lost, which is acceptable for a centered source.

3. Sum all stems; the unassigned residual ($1 - \sum_i \text{masks}$) stays centered.

4. **Level follows position.** On a physical stage a source at center is closer to a center listener than one at the edge, so relocated content is scaled by the inverse-distance gain $g = r(d_0)/r(d_1)$ between its original and target deviations ($\delta = 2$, i.e. +1 dB max center vs. edge, sounds most natural; see the design notes).

5 Expand/contract without decomposition (per-bin ILD warp)

A higher-quality, simpler move that uses no NMF: warp the per-bin ILD directly, $\text{ild}' = \gamma \text{ild}$ (equivalently $r' = r^\gamma$), then renormalize each bin to preserve its energy, $|L'|^2 + |R'|^2 = |L|^2 + |R|^2$. The per-channel gains are real and non-negative, so phase and ICPD are preserved exactly; $\gamma > 1$ expands the stage, $\gamma < 1$ contracts it. γ is solved by bisection so that the 95th percentile of the warped deviation equals half the requested stage width. A original→warped crossfade is a pure amplitude interpolation (same reason), giving an independent “amount” knob. Because neither phase nor spectrum is touched, quality loss is minimal — the main risk is that per-bin ILD fluctuation (interference between overlapping sources) is amplified along with the position. See the design notes for the derivation and listening results.

6 Rotation without decomposition (rigid position shift)

Complementary to the warp: a recording can carry a *global* left/right bias (microphone placement, balance) that expansion cannot fix — a biased image, expanded, only gets more biased. Rotation moves the whole scene rigidly: every bin’s deviation is shifted by a constant, $d' = \text{clip}(d + \theta, \pm 90^\circ)$ — the “rotate the stereo mic” model. The position spread is preserved; only the bias moves. The shift is applied in deviation space and mapped back through the equal-power ILD relation, then re-weighted with the same energy-preserving, phase-preserving gain as the warp, so rotation and expansion share one code path and compose (rotate first, then expand about the corrected center). See the rotation notes for the derivation and measurements.

7 Recoverable vs. invented spread

“Enhancement” means one of two different things, and the tool should support both:

1. **Recoverable spread.** The original has weak-but-real per-activity ILD/ICPD (a true stereo pair, merely centered). The measured cues are *amplified*: target pan = measured pan $\times$ spread factor. This is restoration, and the spatial fingerprint indicates how much real information exists.
2. **Invented spread.** The original is truly mono or hard-centered (ILD/ICPD ≈ 0 everywhere); there is nothing to recover. The tool then acts as a *creative remix*: detection supplies the activities, and the user assigns positions from scratch.

A real stereo orchestra recording is usually case 1 (some cue exists, especially in the mids). A good v1 therefore displays the measured fingerprint and lets the user either “spread measured $\times 2$ ” or set an absolute pan.

8 Minimal v1 pipeline

1. A centered stereo orchestra recording.
2. STFT of both channels (long window, $\approx 75\%$ overlap).
3. NMF with shared D , rank ≈ 8 –16, separate A_L / A_R .
4. Per component: spectrum + activation + ILD histogram $\Rightarrow$ user assigns a pan.
5. Mono stem $\rightarrow$ equal-power pan $\rightarrow$ sum $\rightarrow$ A/B against the original.

9 Open questions

- How separable are the activities at rank 8–16 on real recordings? This determines whether the user gets a clean short list or a muddy one. *Validate early.*
- Add envelope/transient features (attack/decay shape) to the clustering, beyond raw spectrum.
- Constant pan per activity vs. frequency-dependent (per-band) panning.
- Evaluation: A/B listening plus per-band ILD distributions before/after.

10 References

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3. V. Pulkki, “Virtual sound source positioning using vector base amplitude panning,” *J. Audio Eng. Soc.*, vol. 49, no. 10, Oct. 2001. Vector-based reformulation of amplitude panning; the equal-power pan law used by the rendering stage.